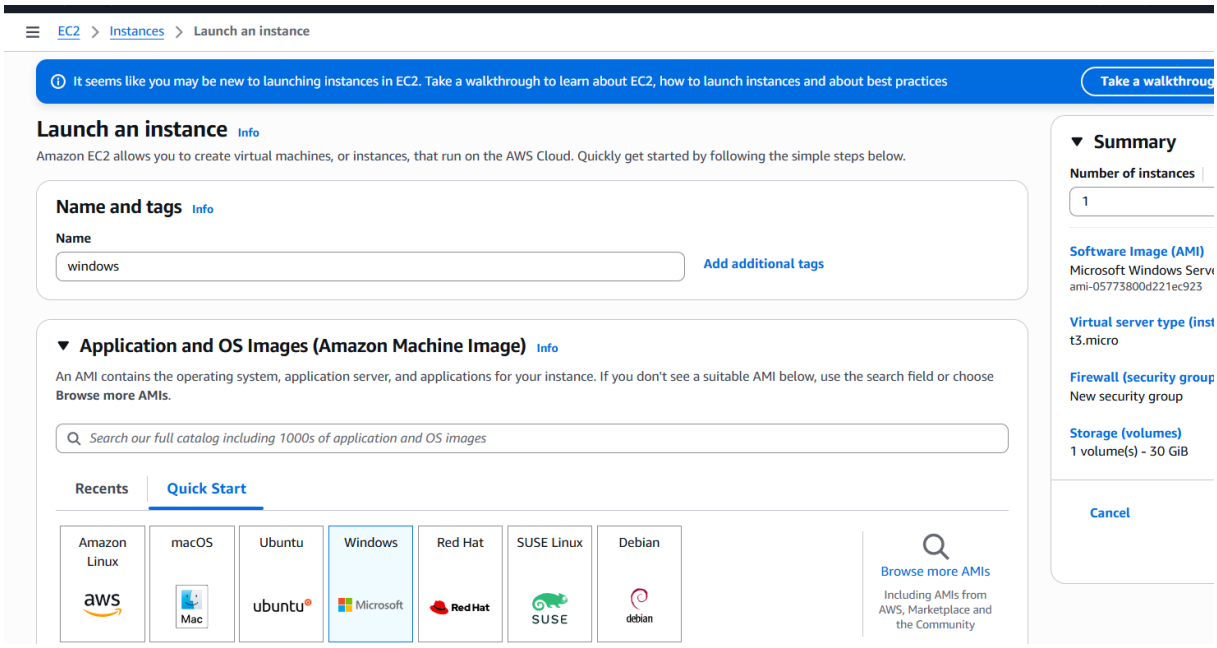
EBS VOLUME IN EC2 (WINDOWS)

An EBS volume is a block storage device used with Amazon Web Services EC2 instances. It provides persistent storage that can be attached to a virtual server.

HANDS-ON FOR EBS VOLUME IN EC2 WINDOWS.

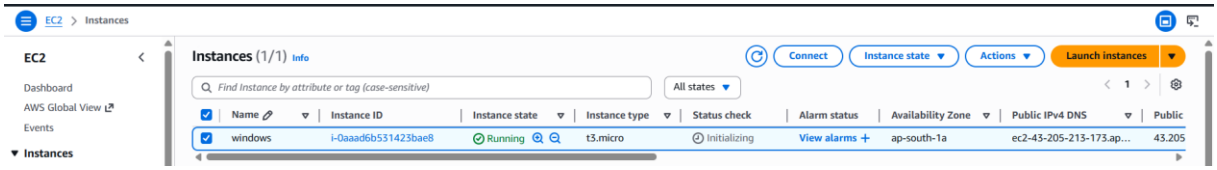
Screenshot: Creating Windows EC2 Instance

- Entered the instance name
- Selected Windows operating system



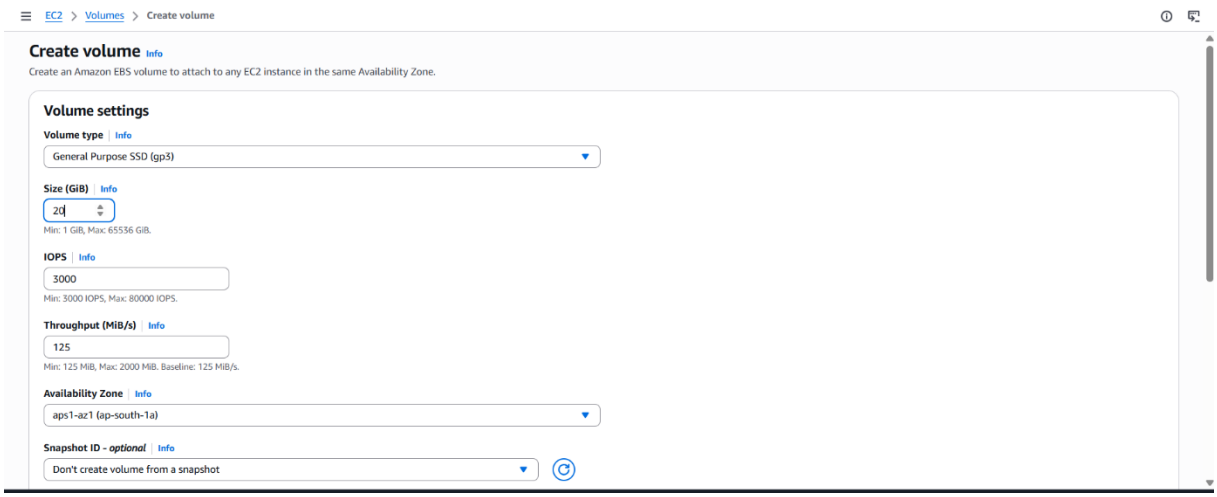
Screenshot: Created Windows EC2 Instance

- Windows instance created successfully.



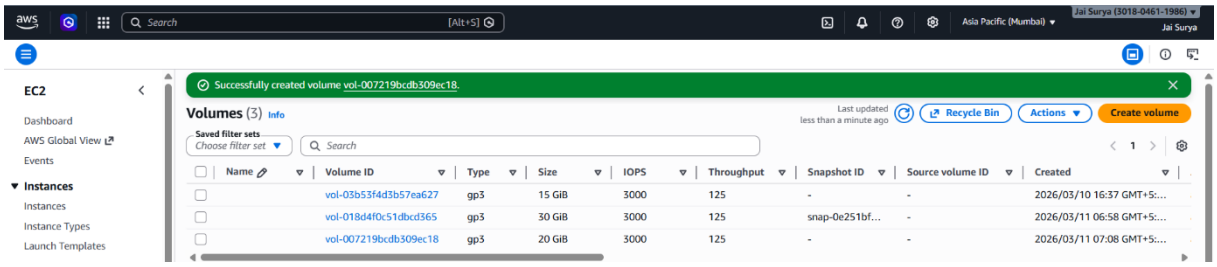
Screenshot: Creating EBS Volume

- Navigated to Elastic Block Store → Volumes
- Clicked Create Volume
- Selected size and availability zone same as EC2 instance



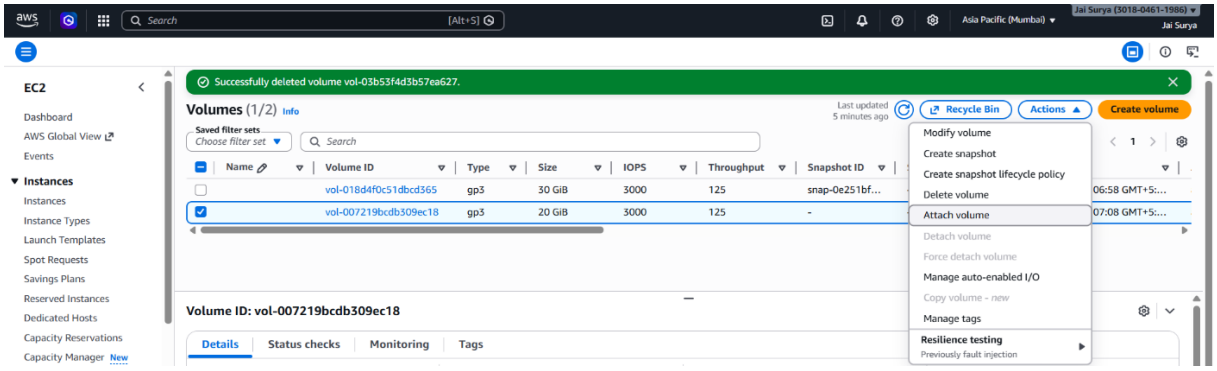
Screenshot: Volume Created

- EBS volume created successfully
- Volume state shows Available



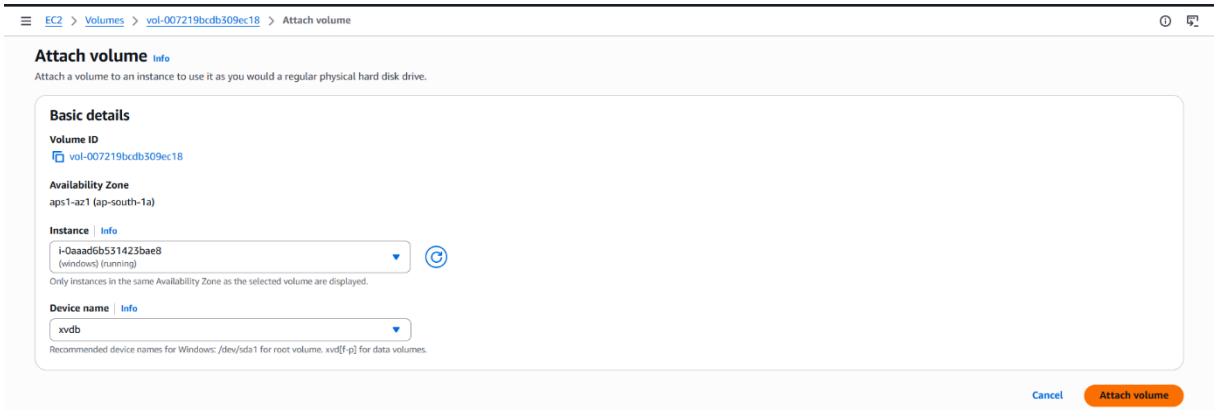
Screenshot: Attaching Volume

- Selected the EBS volume
- Clicked Actions → Attach Volume



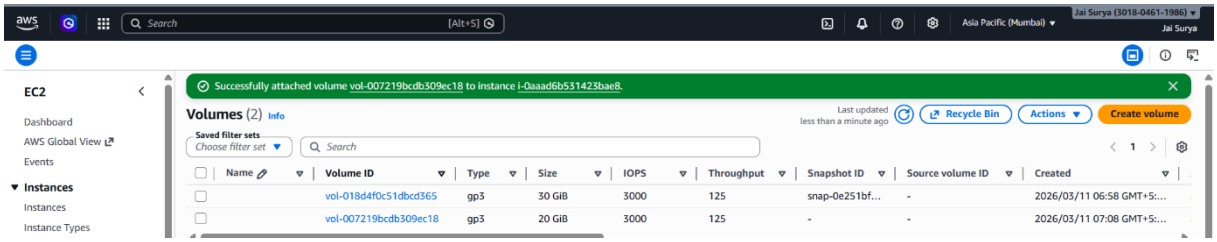
Screenshot: Selecting Instance and Device Name

- Selected the EC2 instance to attach the volume
- Entered the device name



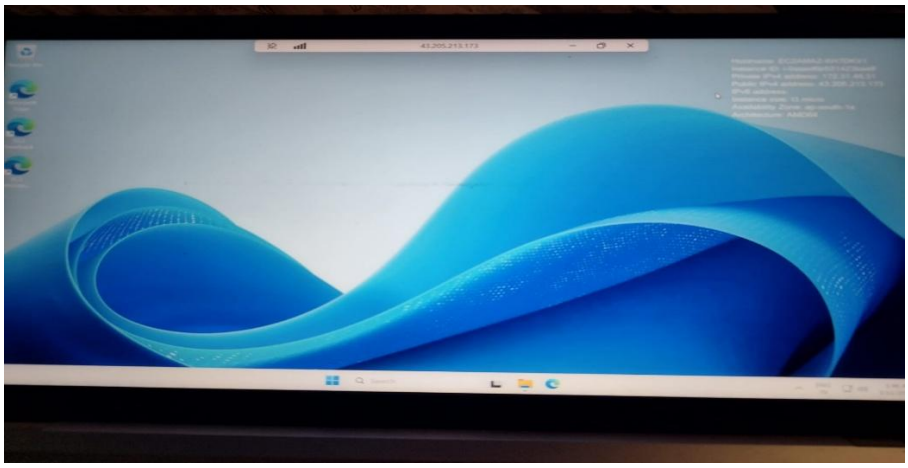
Screenshot: EBS Volume Attached

- EBS volume is successfully attached
- Volume state changes to In-use



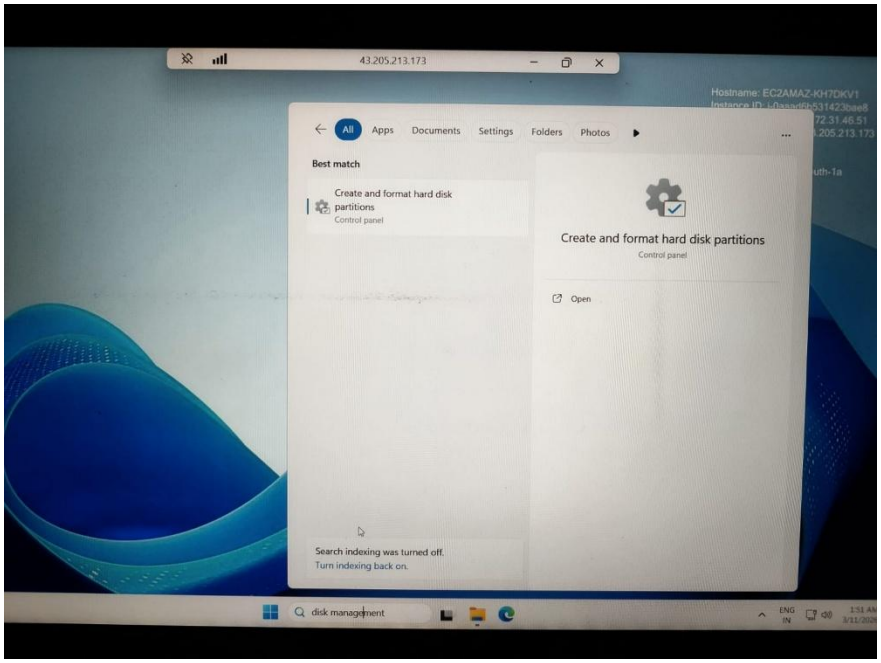
Screenshot: Connecting to Windows EC2 Instance

- Selected the Windows EC2 instance
- Clicked Connect
- Chose RDP Client and downloaded the Remote Desktop file



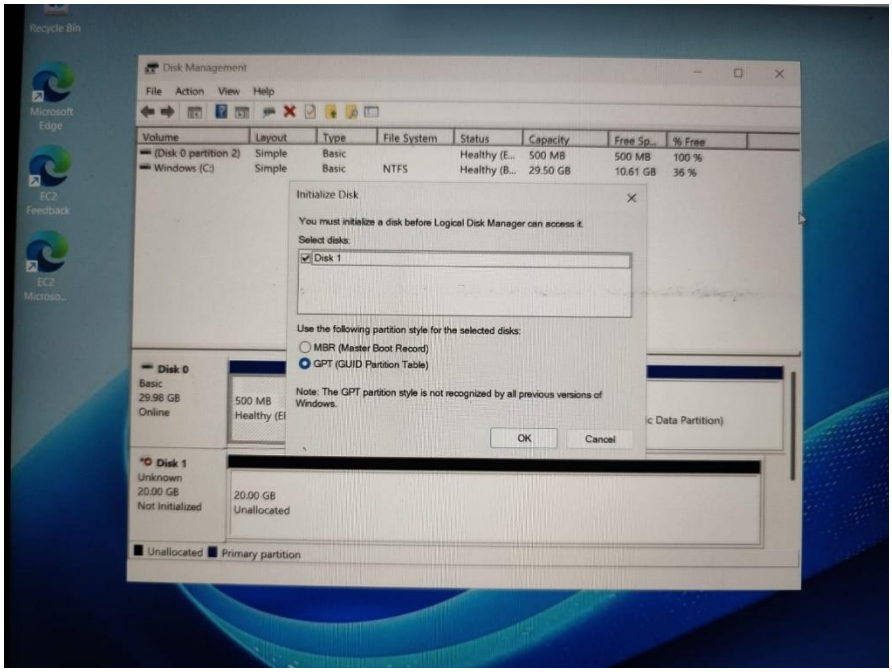
Screenshot: Opening Disk Management

- Clicked the Search bar in the Windows instance
- Searched for Create and format hard disk partitions



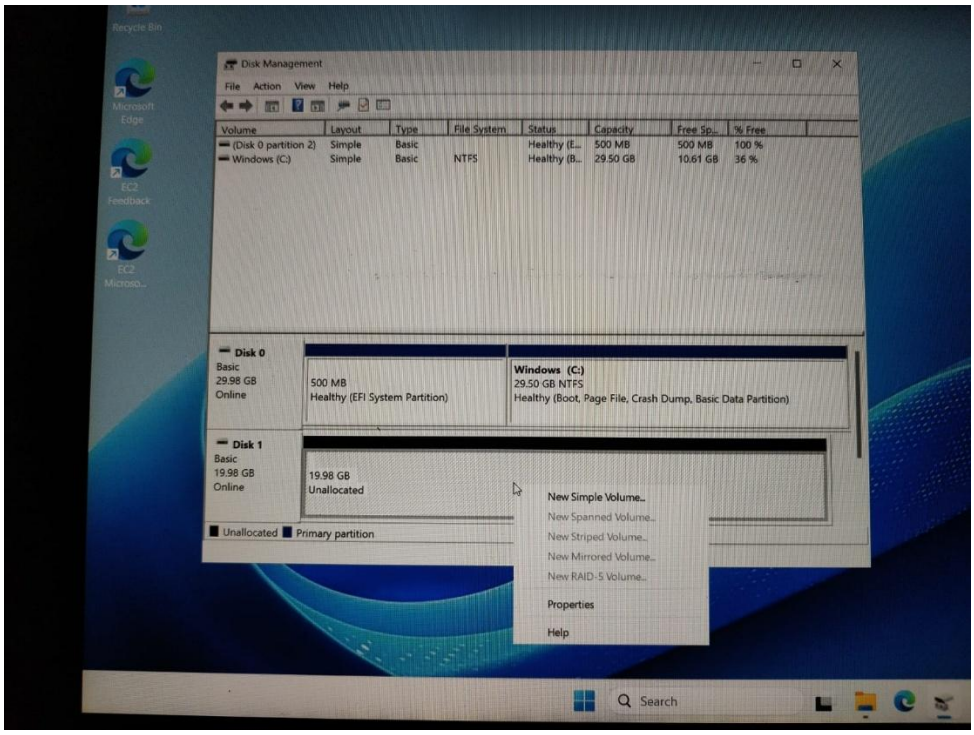
Screenshot: Initialize Disk

- The Initialize Disk window appeared in Disk Management
- Selected the disk and kept the default partition style
- Clicked OK to initialize the disk.



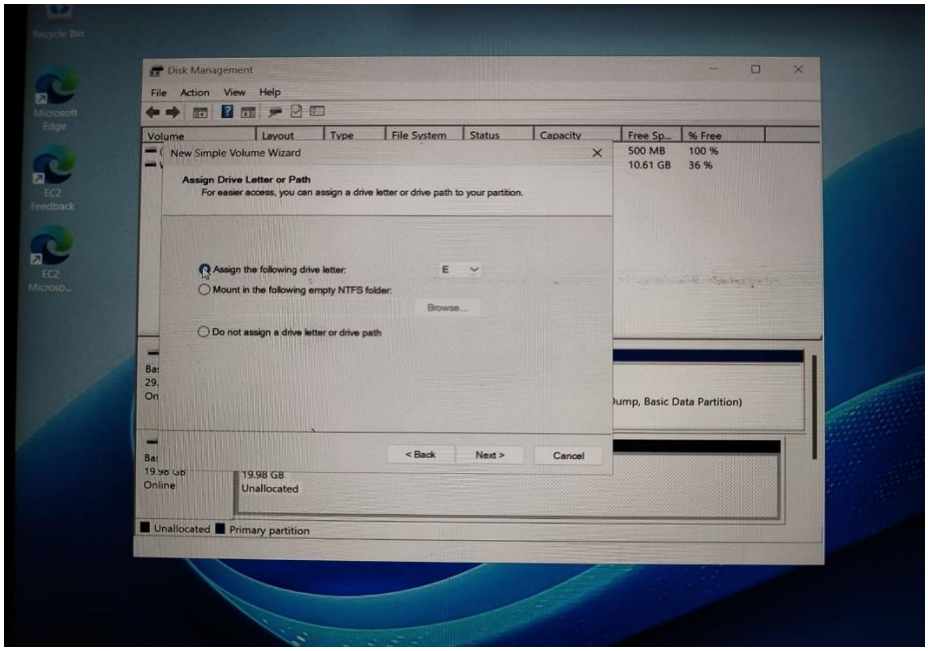
Screenshot: Creating New Simple Volume

- Right-clicked on the Unallocated disk space
- Clicked New Simple Volume
- Started the process to create a new volume on the attached disk.



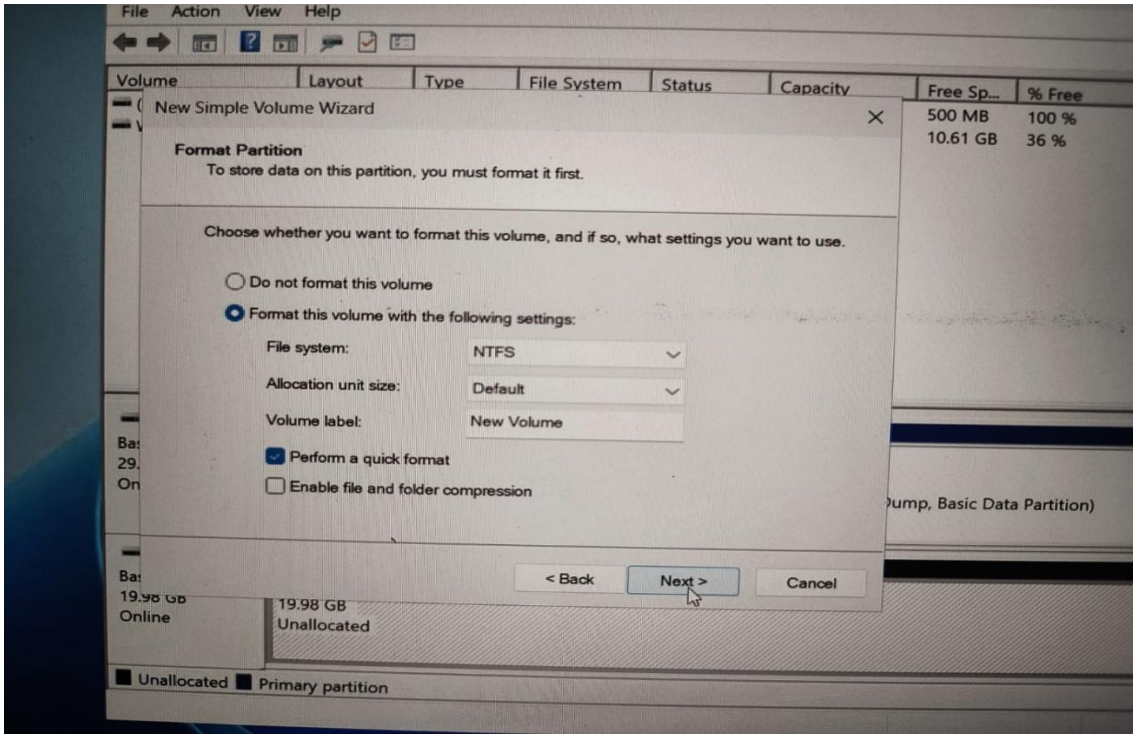
Screenshot: Selecting Drive Letter

- Selected a drive letter for the new volume
- Assigned the drive letter to access the disk in the Windows system.



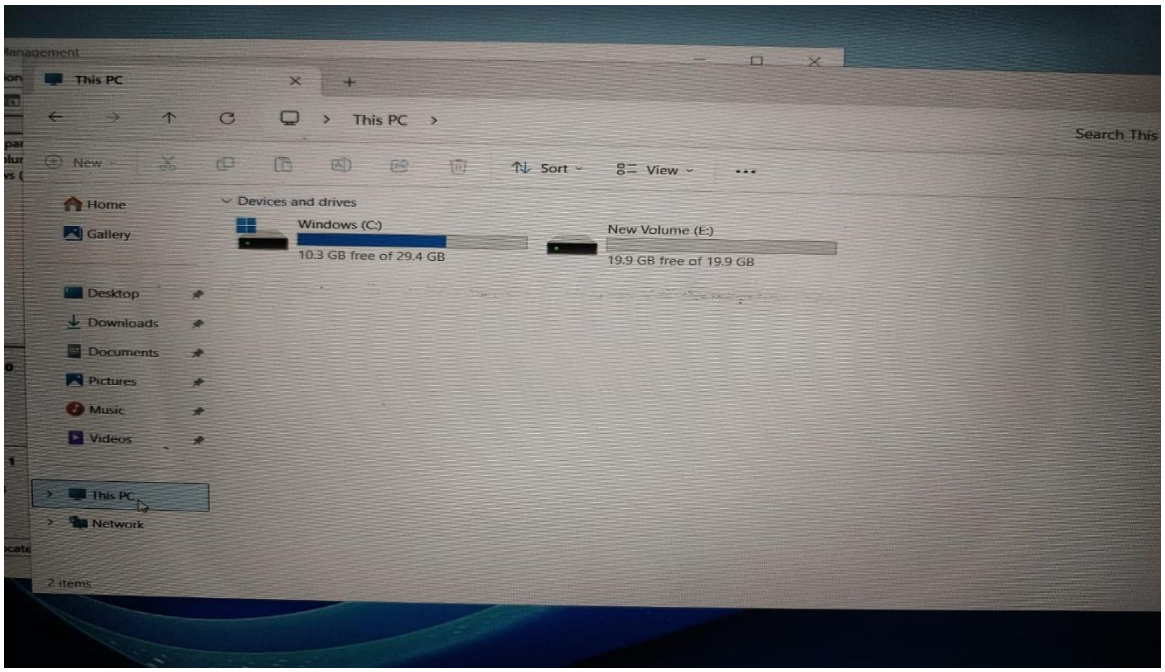
Screenshot: Selecting File System

- Selected the NTFS file system
- Chose the default allocation unit size



Screenshot: Volume Formatted Successfully

- The new volume is formatted successfully
- The disk is now ready to use
- The drive appears in This PC with the assigned drive letter.



SNAPSHOT (EBS SNAPSHOT)

A snapshot is a backup copy of an EBS volume stored in Amazon Web Services. It captures the data of the volume at a specific point in time. Snapshots are mainly used for backup, recovery, and creating new volumes from existing data.

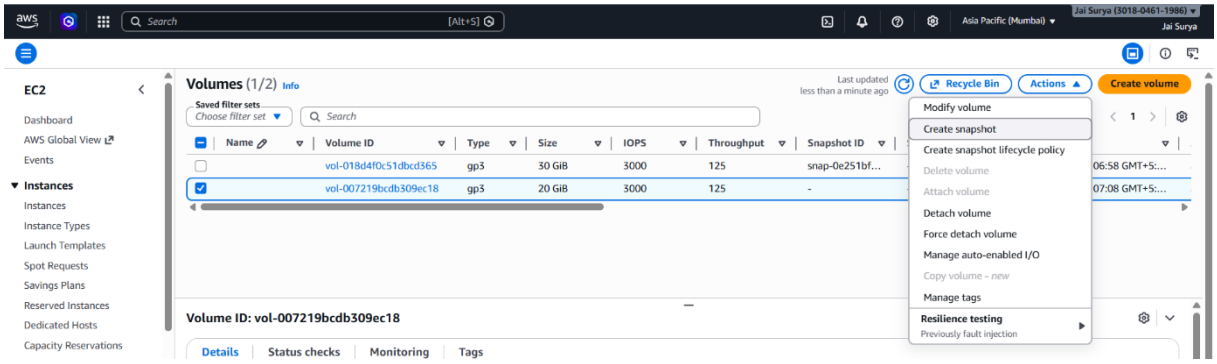
Features of Snapshot

- Provides backup of EBS volumes
- Stored automatically in Amazon S3
- Used to restore data when needed

HANDS-ON FOR SNAPSHOT OF A VOLUME (EBS SNAPSHOT).

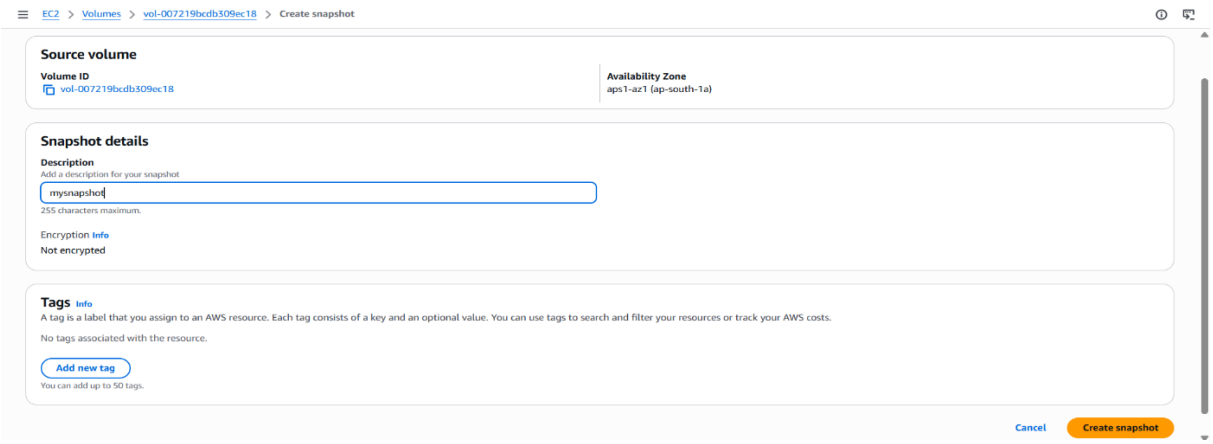
Screenshot: Creating Snapshot from Volume

- Selected the EBS volume
- Clicked Actions
- Selected Create Snapshot



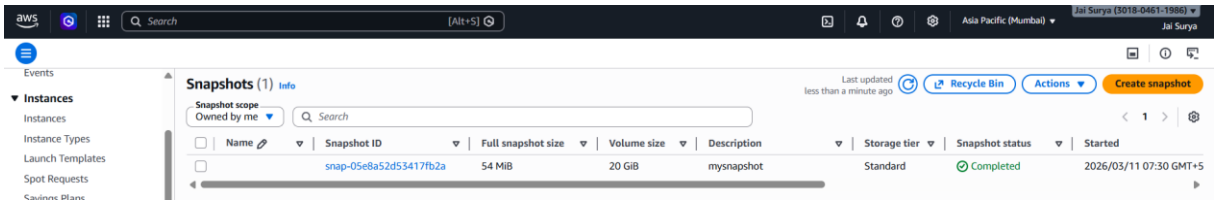
Screenshot: Entering Snapshot Name

- Entered the snapshot name
- Added description if required



Screenshot: Snapshot Created Successfully

- Snapshot creation completed successfully
- Snapshot status shows Completed



AMI (AMAZON MACHINE IMAGE)

An AMI is a template used to create virtual servers (EC2 instances) in Amazon Web Services. It contains the operating system, application server, and required software needed to launch an instance.

An AMI also acts as a backup of an EC2 instance, because it stores the instance configuration and attached storage snapshot. This allows users to recreate the same server whenever needed.

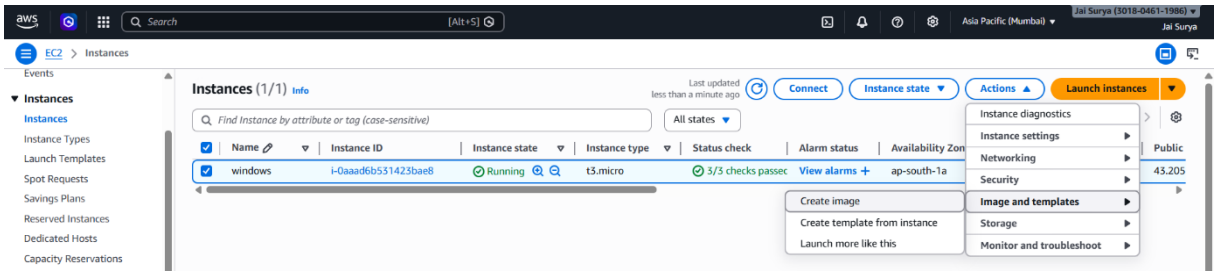
Features of AMI

- Used to launch EC2 instances
- Contains OS and pre-configured software
- Helps in quick deployment of servers
- Can create multiple instances from one AMI

HANDS-ON FOR AMI OF AN INSTANCE

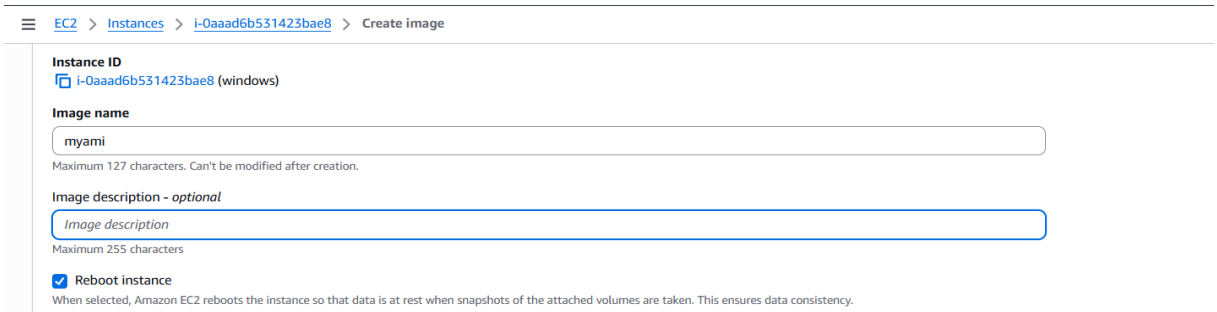
Screenshot: Creating AMI from EC2 Instance

- Selected the EC2 instance
- Clicked Actions → Image and templates → Create Image
- Started the process to create an AMI (backup of the instance)



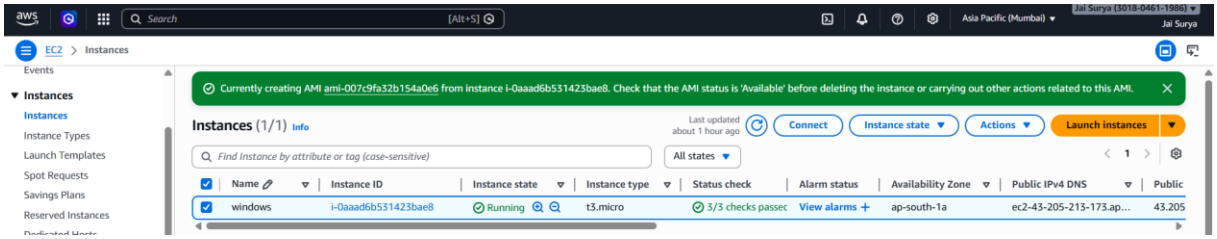
Screenshot: Entering AMI Name

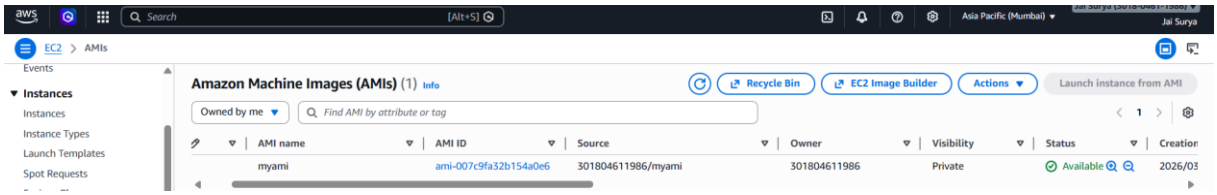
- Entered the AMI name
- Added a description (optional)



Screenshot: AMI Created Successfully

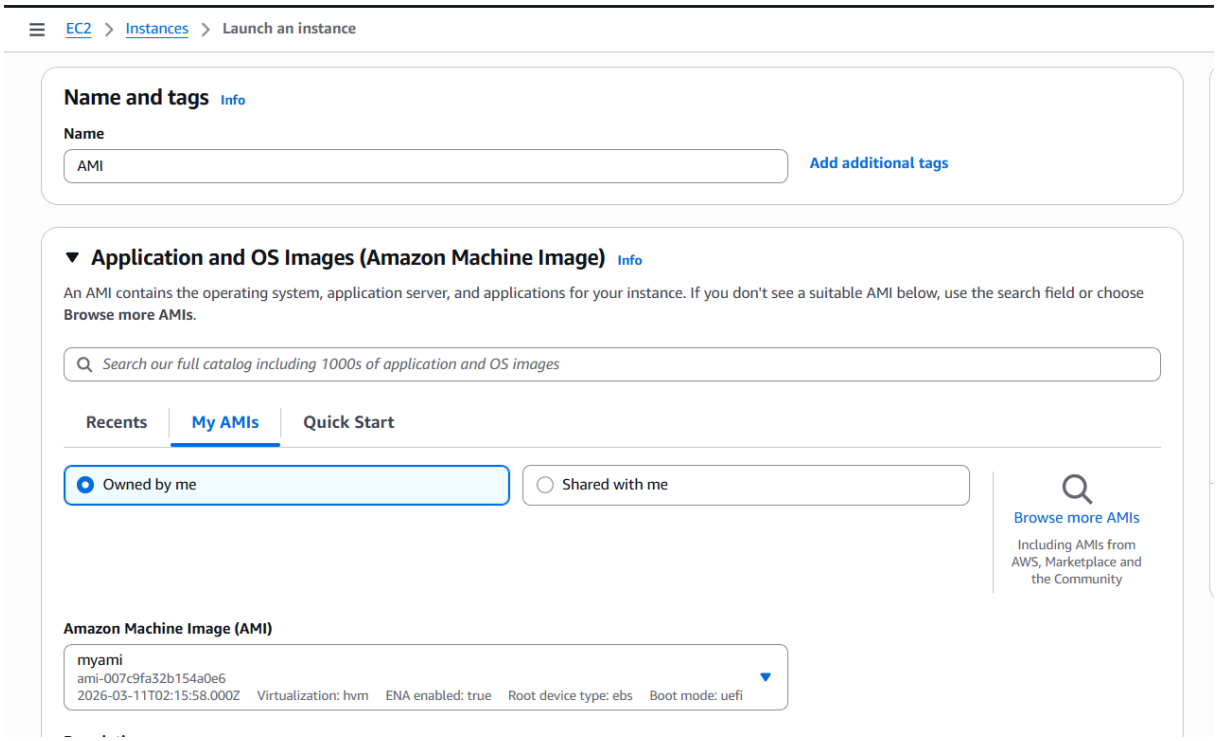
- AMI creation completed successfully
- Status shows Available





Screenshot: Launching EC2 from AMI

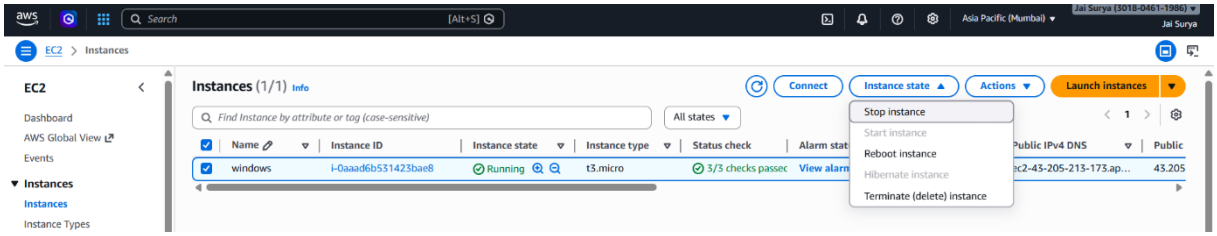
- Navigated to Launch Instance in the EC2 dashboard
- Selected the "My AMIs" tab
- Found the created AMI under "Owned by me"
- Ready to launch a new EC2 instance using the AMI as a template



CHANGING EC2 INSTANCE TYPE AFTER CREATION (LINUX OR WINDOWS):

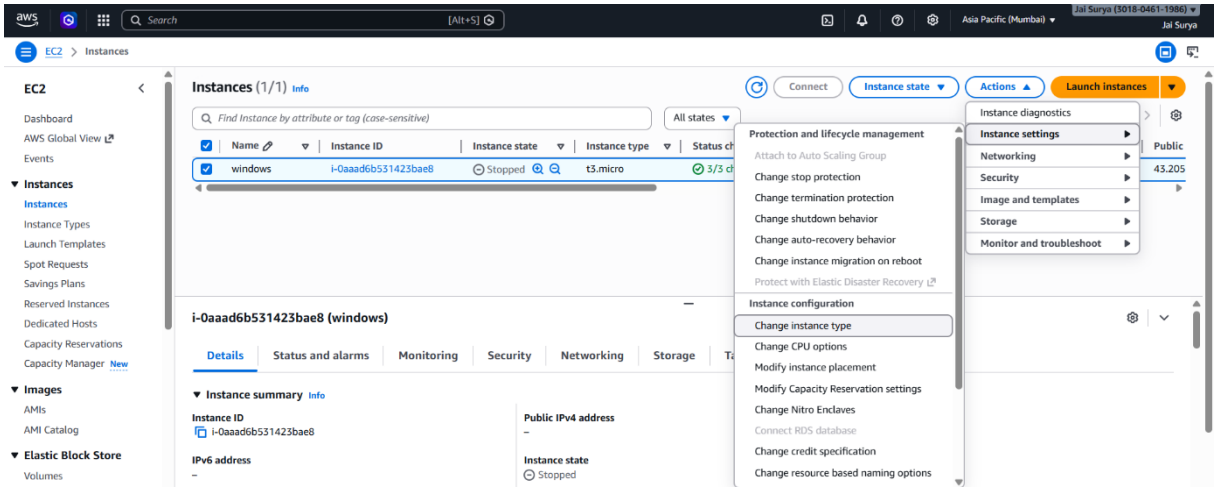
Step 1: Stop the Instance

- Go to the EC2 dashboard
- Select the instance you want to change
- Click Actions → Instance State → Stop
- Confirm to stop the instance (you cannot change instance type while it is running)



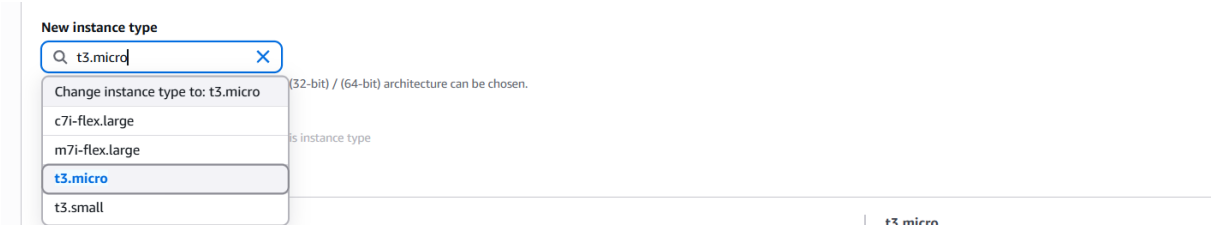
Step 2: Change Instance Type

- With the instance stopped, select the instance
- Click Actions → Instance Settings → Change Instance Type
- Choose the new instance type (e.g., t2.micro → t3.medium)
- Click Apply



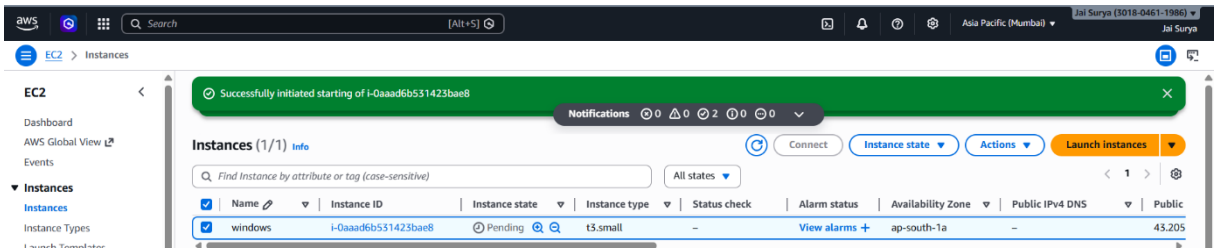
Step 3: Start the Instance

- With the instance still selected, click Actions → Instance State → Start
- Wait for the instance to boot up with the new instance type



Step 4: Verify Instance Type

- Select the instance
- Check the Instance Type column to confirm the change



HTTP VS HTTPS – DIFFERENCES

Feature	**HTTP (Hypertext Transfer Protocol)**	**HTTPS (Hypertext Transfer Protocol Secure)**
Full Form	Hypertext Transfer Protocol	Hypertext Transfer Protocol Secure
Security	Not secure, data is sent in plain text	Secure, data is encrypted using SSL/TLS
Port	80	443
Encryption	No encryption	Uses SSL/TLS for encryption
Data Integrity	Vulnerable to interception and tampering	Ensures data integrity
Authentication	No authentication	Provides server authentication via SSL certificate
Use Case	Suitable for **non-sensitive data**	Recommended for **sensitive data** like login, payment, personal info